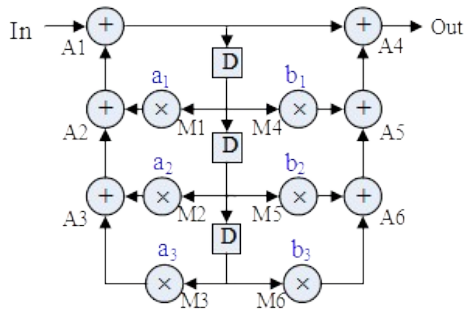


2018 Spring VLSI DSP Homework Assignment #5

For the design shown below.



Part A. (due date: 2018/6/5)

- a) For a folded design with 4 folding sets (SA1, SA2, SM1, SM2,) and a folding factor equal to 3, verify if this is a valid folding. Assume Multiplier is 2-stage pipelined and adder is 1-stage pipelined.

SA1={A3,A2,A1}	SM1={M3,M2,M1}
SA2={A6,A5,A4}	SM2={M6,M5,M4}

- b) If it is not, find a new valid folding design by modifying the folding orders in each set.
Note: Each folding set has 6 possible folding orders. In total, there are $6^4=1296$ possible combinations of folding orders. You may write a program to enumerate all 1296 possible combinations and check if any of them are valid.
- c) In case, you cannot identify any valid combinations of folding orders in (b), perform retiming first followed by folding
- d) Derive your folded design without register minimization based on the folding obtained in (b) or (c). You need to draw the folded circuit design.

Part B. (due date: 2018/6/12)

- e) Repeat (d) and use the forward backward register allocation scheme to minimize the register usage. **Note:** you need to show the derivation details, e.g. life time chart, register allocation table, and so on.
- f) Write Verilog designs for both the original design and the folded design in (e) Use Verilog simulation results to verify your folded design is correct. Assume inputs are 8-bit wide signed integers. Multiplier coefficients are all 8-bit wide and have magnitudes between 0.5 and 1. You may determine the word length of other variables so that no overflow occurs.